

Yersinia pseudotuberculosis Infectious Disease — Comprehensive Disease-Characteristics Report

MONDO ID: MONDO:0007024 | **Category:** Infectious Disease | **Causative agent:** *Yersinia pseudotuberculosis* (NCBI:txid633)

Evidence base: Aggregated disease-level literature (PubMed reviews, primary microbiology/immunology studies, human clinical case series, model-organism experiments). No individual patient/EHR data or omics files were provided; all statements are literature-derived and cited by PMID.

Summary

Yersinia pseudotuberculosis infectious disease is an acute, zoonotic, foodborne enteric infection caused by the cold-tolerant (psychrotrophic) Gram-negative coccobacillus *Y. pseudotuberculosis*. The pathogen is acquired fecal-orally from contaminated water, produce, or meat, and it uniquely proliferates under refrigeration, which shapes both its epidemiology and its prevention. The disease shows a striking **geographic dichotomy**: in Europe and most of the world it manifests as a self-limiting gastroenteritis or, in children, as fever with mesenteric lymphadenitis and terminal ileitis that mimics acute appendicitis ("pseudoappendicitis"). In Russia and Japan, by contrast, epidemic strains cause a severe systemic inflammatory disease historically called **Far East scarlet-like fever (FESLF)**, characterized by scarlatiniform rash, arthralgia, and toxic-shock-like features.

The pathophysiology is bacterial-effector driven, not host-genetic. The organism enters through M cells over the Peyer's patches using the chromosomal adhesin **invasin** (a high-affinity β 1-integrin ligand), replicates in the intestine, and then disseminates to mesenteric lymph nodes, liver, and spleen. A **temperature-regulated type III secretion system (T3SS)** encoded on the pYV virulence plasmid — switched on at 37 °C by the master regulator **LcrF** — injects **Yop effector proteins** that paralyze phagocytes and disable innate immune signaling (NF- κ B, MAPK, IRF3). The severe FESLF phenotype is additionally determined by strain

plasmid/superantigen content: the **pVM82 plasmid** and the superantigen **YPMa (Y. pseudotuberculosis-derived mitogen A)**, which overstimulates CD4⁺ Vβ3/Vβ7/Vβ8 T cells to release a TNF-α/IFN-γ cytokine storm.

The disease is generally self-limiting with an excellent prognosis, and the organism is broadly antibiotic-susceptible (fluoroquinolones, third-generation cephalosporins, aminoglycosides, tetracyclines). Severe systemic and septicemic disease is concentrated in hosts with **iron overload, desferrioxamine therapy, or immunosuppression**. Post-infectious immune sequelae — reactive arthritis and erythema nodosum — occur in a minority. Notably, *Y. pseudotuberculosis* is the recent enteric evolutionary ancestor of *Y. pestis* (plague). Prevention rests on food and water hygiene; no human vaccine exists. This report synthesizes 12 confirmed findings from 46 reviewed papers across all 15 requested disease-characteristic domains.

1. Disease Information

Overview. *Yersinia pseudotuberculosis* infectious disease (yersiniosis due to *Y. pseudotuberculosis*) is an acute enteric zoonosis. Clinically it presents along a spectrum from self-limiting gastroenteritis to pseudoappendicitis (mesenteric adenitis/terminal ileitis) to, in the Far East, a severe systemic scarlet-fever-like illness. It is caused by a Gram-negative, facultatively anaerobic, motile (at ≤30 °C), psychrotrophic member of the family *Yersiniaceae*.

Key identifiers: - **MONDO:** MONDO:0007024 - **MeSH:** *Yersinia pseudotuberculosis* Infections - **ICD-10:** A28.2 (Extraintestinal yersiniosis) / A04.8 (other bacterial intestinal infections) - **Pathogen taxonomy:** NCBI:txid633 (*Yersinia pseudotuberculosis*) - OMIM/Orphanet: not a Mendelian disorder; no OMIM entry (infectious, not genetic). Orphanet lists Far East scarlet-like fever as a rare condition.

Synonyms / alternative names: Far East scarlet-like fever (FESLF); Izumi fever (historical Japanese name); pseudotuberculosis; Pasteurella pseudotuberculosis infection (historical bacterial name); scarlatinoid fever.

Information source type. The evidence base is **aggregated disease-level literature** (case series, outbreak investigations, microbiology and animal-model studies, and serosurveys) rather than individual EHR-derived cohorts. Much quantitative sequelae data are extrapolated from the closely related *Y. enterocolitica*.

The two-faced clinical identity is the anchoring finding (**F001**):

"Far East scarlet-like fever is caused by Yersinia pseudotuberculosis infection, an organism that typically causes self-limiting gastroenteritis in Europe." — Amphlett 2016, PMID: 26819960

"Geographical heterogeneity exists between virulence factors produced by European and Far Eastern Y pseudotuberculosis strains, implicating superantigen Y pseudotuberculosis-derived mitogen A (YPMa) in the pathogenesis of FESLF." — PMID: 26819960

2. Etiology

Primary cause. The disease is **infectious**, caused entirely by *Y. pseudotuberculosis*. There is no genetic (Mendelian) etiology in the human host. Disease severity is governed by **pathogen genotype** (plasmid and superantigen content) interacting with **host iron status and immune competence**.

Environmental / exposure risk factors: - **Foodborne/waterborne transmission** via contaminated refrigerated food, raw vegetables (lettuce, carrots), untreated water, and meat (F012). - **Animal contact** — broad wildlife and domestic-animal reservoirs (F008). - **Age** — children are disproportionately affected by the pseudoappendicitis form. - **Cold storage** — the organism's psychrotrophic growth increases the infectious dose reachable in refrigerated foods.

Host risk factors for severe/systemic disease (F010): - **Iron overload** (e.g., β -thalassemia, hemochromatosis). - **Desferrioxamine (DFO) iron-chelation therapy** — DFO acts as a xenosiderophore delivering iron to the bacterium. - **Immunosuppression and splenectomy**.

"Patients undergoing DFO therapy are at risk for Y. enterocolitica infection which may be localized to mesenteric nodes and tonsils or occur as a generalized form such as septicemia." — Wanachiwanawin 2000, PMID: 11132234

Genetic protective / risk factors: None established in humans. Susceptibility is essentially universal; outcome modulation is by iron availability and immune status rather than host germline variants.

Gene–environment interaction. The key "interaction" is **host iron availability × bacterial iron acquisition**: excess free iron (from overload or DFO chelation therapy) supports rapid bacterial growth and systemic dissemination, converting a normally contained enteric infection into septicemia.

3. Phenotypes

Phenotype	Type	Characteristics	Frequency	Suggested HPO
Fever	Symptom	Acute onset; prominent in FESLF	Very common	HP:0001945
Abdominal pain (RLQ) / pseudoappendicitis	Sign/symptom	Mesenteric adenitis, terminal ileitis; children 5–14 y	Common in children	HP:0002027, HP:0002605
Diarrhea / gastroenteritis	Symptom	Self-limiting (Europe)	Common	HP:0002014
Mesenteric lymphadenitis	Clinical sign	Mimics appendicitis	Common (children)	HP:0004299
Scarlatiniform rash	Physical manifestation	FESLF hallmark	Far Eastern strains	HP:0000988
Arthralgia / reactive arthritis	Sign; post-infectious	Weeks after infection	~12% (Yersinia sequele est.)	HP:0002829, HP:0001369
Erythema nodosum	Physical; post-infectious	Panniculitis, self-limiting	~3% (Yersinia sequele est.)	HP:0012219
Transient renal dysfunction	Lab/clinical	Systemic FESLF; correlates with anti-YPM	Systemic cases	HP:0000083
Toxic-shock-like features	Clinical	Superantigen-driven	Severe FESLF	HP:0032169

Phenotype spectrum evidence (F006):

"responsible for scarlatinoid fever, food poisoning, post-infectious complications like erythema nodosum/reactive arthritis as well as pseudoappendicitis in children" — Basharat 2021, [PMID: 33321204](#)

Reactive-arthritis frequency (extrapolated from *Y. enterocolitica* population study, F006):

"Self-reported symptoms consistent with ReA were reported by 12% of yersiniosis patients compared to 5% in a reference group not exposed to yersiniosis." — Rosner 2013, [PMID: 23701958](#)

Onset/severity/progression: acute onset; self-limiting in most; systemic FESLF is more severe and correlates with higher anti-YPM titers. **Quality of life:** acute illness is generally short-lived; the main QoL burden falls on the minority with reactive arthritis (weeks–months of joint pain).

4. Genetic/Molecular Information

This is an infectious disease; there are no human causal genes, pathogenic germline variants, chromosomal abnormalities, or Mendelian inheritance. The "genetics" of the disease are the genetics of the **pathogen**.

Key bacterial virulence loci: - pYV / pCD1 virulence plasmid (~70 kb): encodes the Ysc T3SS and Yop effectors (YopH, YopE, YopT, YopJ/YopP, YopM, YopO/YpkA) plus the master regulator LcrF (VirF). - Chromosomal *inv* gene: * *invasin*, the β 1-integrin adhesin for M-cell entry. - *ail*, *yadA* (plasmid): *adhesion/serum resistance*. - *YPMa* gene (*ypmA**): superantigen; hallmark of Far Eastern strains. - pVM82 plasmid (~82 MDa): present only in FESLF-causing strains (F011).

Strain plasmid genotype determines clinical severity (F011):

"effects of pathogenicity of an understudied pVM82 plasmid present only in Y pseudotuberculosis strains causing clinical-epidemic manifestation of the infections as Far East scarlet-like fever (FESLF)" — Somova 2016, PMID: 30695393

"Variability of damage of innate immunity cells and target-organs caused by various plasmid types of Y pseudotuberculosis by virulence could determine polymorphism of clinical-morphological manifestations of this infection." — PMID: 30695393

Serotypes: ≥21 O-serotypes exist; *Y. pestis* evolved from serotype **O:1b (F005)**.

5. Environmental Information

Infectious agent (NCBI Taxon): *Yersinia pseudotuberculosis*, NCBI:txid633.

Environmental/transmission factors: - **Cold tolerance (psychrotrophy):** enables growth in refrigerated foods (**F012**).

"Yersinia enterocolitica and Yersinia pseudotuberculosis are important foodborne pathogens that cause infections through contaminated refrigerated food." — Palonen 2010, PMID: 20088683

- **Food vehicles:** raw vegetables, lettuce, carrots, untreated water, meat.
- **Animal reservoirs:** pigs, sheep, wild birds, rodents, deer, and many others; food/livestock strains are molecularly indistinguishable from human isolates (**F008**).

"All human Y. pseudotuberculosis 1/O:1 strains were indistinguishable from pig, sheep or food strains." — Okwori 2009, PMID: 19835774

Lifestyle factors: consumption of raw/undercooked produce and unpasteurized/untreated water; exposure to farm and wild animals.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation)

1. Ingestion of contaminated refrigerated food/water
→ delivers viable *Y. pseudotuberculosis* to the small intestine
2. Chromosomal invasin binds host $\beta 1$ -integrins on M cells (Peyer's patches)
→ results in translocation across the follicle-associated epithelium
3. Invasin- $\beta 1$ -integrin engagement activates Rac1, MAP kinases, NF- κ B
→ drives local chemokine production and bacterial uptake
4. Bacteria replicate in the intestinal lumen/lamina propria
→ establishes a replicating pool (required for later dissemination)
5. Host body temperature (37 °C) is sensed by an RNA thermometer + YmoA
→ de-represses the master regulator LcrF
6. LcrF activates the pYV-encoded Ysc T3SS and yop genes
→ injects Yop effectors into phagocytes on contact
- 7a. YopH/YopE/YopT/YopO disrupt phagocyte cytoskeleton/Rho GTPases
→ results in resistance to phagocytosis (antiphagocytic defense)
- 7b. YopJ acetylates/deubiquitinates TAK1 and reduces K63-ubiquitination of TRAF3/TRAFF6
→ blocks NF- κ B, MAPK, and IRF3 signaling
→ suppresses innate cytokine and interferon responses
8. Surviving extracellular bacteria disseminate to mesenteric lymph nodes, liver, spleen
→ mesenteric lymphadenitis / pseudoappendicitis / abscessation

BRANCH: strain genotype	
European strains (YPMa-/pVM82-)	Far Eastern strains (YPMa+, pVM82+)
→ localized, self-limiting gastroenteritis / pseudoappendicitis	→ superantigen released

9. (Far East) YPMa superantigen cross-links MHC-II to TCR V β 3/V β 7/V β 8
→ massive polyclonal CD4+ T-cell activation
10. Activated T cells release TNF- α and IFN- γ
→ systemic inflammation, rash, renal dysfunction, toxic shock (FESLF)
11. (Post-infectious, in a minority) molecular mimicry / immune complexes
→ reactive arthritis and erythema nodosum weeks after infection

Detail by category

Molecular pathways. Host invasin signaling engages **Rac1 → MAPK → NF- κ B**. Bacterial YopJ targets the **NF- κ B, MAPK, and IRF3** pathways (**F009**):

"YopJ inhibited TLR-mediated NF-kappaB and MAP kinase activation, as suggested by previous studies. In addition, induction of the TLR-mediated interferon response was blocked by YopJ, indicating that YopJ also inhibits IRF3 signalling." — Sweet 2007, [PMID: 17608743](#)

Cellular processes. Inflammation (GO:0006954), inhibition of phagocytosis (GO:0006909), suppression of innate immune signaling, superantigen-driven T-cell proliferation (GO:0042110).

Protein dysfunction (bacterial effectors as toxins): YopH (tyrosine phosphatase), YopE/YopT (Rho GAP/protease), YopJ (acetyltransferase/deubiquitinase), YopO/YpkA (kinase), YopM (leukocyte-modulating). LcrF is the thermo-activated transcriptional master switch.

Temperature control is the master mechanism (F003):

"two different Y. pseudotuberculosis patient isolates expressing a stabilized thermometer variant were strongly reduced in their ability to disseminate into the Peyer's patches, liver and spleen and have fully lost their lethality" — Böhme 2012, [PMID: 22359501](#)

"Ysc-T3SS-mediated Yop secretion leads to global reprogramming of the Yersinia transcriptome with a massive shift of the expression from chromosomal to virulence plasmid-encoded genes" — Meyer 2024, [PMID: 39159284](#)

Intestinal entry mechanism (F004):

"Invasin protein is a high-affinity ligand for beta1 integrins and especially important in the early phase of intestinal infection for efficient translocation through the M cells located in the follicle-associated epithelium overlying the Peyer's patches." — Grassl 2003, [PMID: 12755365](#)

"Replication of bacteria in the intestine before translocation appeared critical for dissemination" — Barnes 2006, [PMID: 16754724](#)

Superantigen mechanism (F002):

"Patients with systemic symptoms such as lymphadenopathy, transient renal dysfunction, and arthritis had significantly higher titers of anti-YPM than patients with gastrointestinal tract symptoms alone." — Abe 1997, PMID: 9109426

"This shock was blocked by the injection of monoclonal antibodies (mAbs) to CD4, TCR Vbeta7 plus Vbeta8, tumor necrosis factor-alpha (TNF-alpha) and interferon-gamma (IFN-gamma)" — Miyoshi-Akiyama 1997, PMID: 9159409

Quantitatively (**F002**): 20/33 (61%) of acutely infected patients had elevated anti-YPM IgG vs controls (P=0.0001); Vβ3⁺ T cells were significantly increased in the acute phase (P=0.009).

Immune involvement: the disease is a contest between bacterial innate-immune evasion (Yop-mediated) and, in FESLF, pathological adaptive over-activation (superantigen-mediated).

Cell types (CL): M cells, macrophages/neutrophils (CL:0000235, CL:0000775), CD4⁺ T cells (CL:0000624), dendritic cells (CL:0000451). **GO biological processes:** GO:0006909 (phagocytosis), GO:0042110 (T-cell activation), GO:0006954 (inflammatory response), GO:0043123 (positive regulation of NF-κB), GO:0009266 (response to temperature stimulus).

7. Anatomical Structures Affected

Organ level (primary): - Terminal ileum and cecum (UBERON:0002116; UBERON:0001153) - Mesenteric lymph nodes (UBERON:0002509) - Peyer's patches / gut-associated lymphoid tissue (UBERON:0003454)

Secondary organ involvement (dissemination/complications): - Liver (UBERON:0002107) and spleen (UBERON:0002106) — hepatosplenic abscessation - Skin (UBERON:0002097) — rash, erythema nodosum - Joints (UBERON:0000467 articular system) — reactive arthritis - Kidney (UBERON:0002113) — transient renal dysfunction in FESLF - Rarely muscle (myositis) and heart (myocarditis)

Body systems: digestive, lymphatic/immune, integumentary, musculoskeletal, and (in systemic disease) renal.

Tissue/cell level: intestinal follicle-associated epithelium (M cells), lymphoid tissue, professional phagocytes (macrophages, neutrophils), CD4⁺ T lymphocytes.

Subcellular (GO Cellular Component): host plasma membrane (GO:0005886, invasin–integrin contact and T3SS translocon), cytosol (GO:0005829, Yop effector site of action), and the bacterial T3SS injectisome.

Localization/lateralization: RLQ abdominal involvement (ileocecal) is characteristic; disease is systemic rather than lateralized.

8. Temporal Development

- **Onset:** acute to subacute after a short incubation (typically days). Predominantly affects **children and young adults** for the pseudoappendicitis form.
- **Course:** most cases are **self-limiting** over 1–3 weeks. FESLF is a more severe acute systemic illness.
- **Stages:** (1) intestinal colonization/replication → (2) lymphatic dissemination (mesenteric adenitis) → (3) possible systemic spread (hepatosplenic; FESLF) → (4) post-infectious immune sequelae (reactive arthritis, erythema nodosum) weeks later.
- **Progression rate:** generally self-limited; severe/septicemic progression is concentrated in iron-overloaded/immunosuppressed hosts.
- **Remission:** spontaneous in most; antibiotic-responsive in systemic disease.
- **Critical period:** the **37 °C temperature shift** on host entry is the decisive window that activates the T3SS virulence program (**F003**).

9. Inheritance and Population

Inheritance: Not applicable — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, founder effects, or carrier frequency.

Epidemiology: - Sporadic worldwide; **epidemic FESLF** clusters in the Russian Far East and Japan (**F011**).

"Pseudotuberculosis in humans until the 1950s was found in different countries of the world as a rare sporadic disease that occurred in the form of acute appendicitis and mesenteric lymphadenitis." — Somova 2020, [PMID: 32498317](#)

- **Geographic distribution:** European strains cause mild disease; Far Eastern strains (YPMa⁺, pVM82⁺) cause severe systemic disease — a genotype-driven geographic virulence gradient (F001).
- **Age distribution:** skewed toward children/young adults for pseudoappendicitis.
- **Sex ratio:** no strong, well-established skew for the enteric form.
- **Serotype geography:** ≥21 O-serotypes; distinct serotype/plasmid profiles by region.

10. Diagnostics

Two-tier diagnostic strategy (F007):

Acute disease — direct detection: - **Stool culture** (cold enrichment exploits psychrotrophy), tissue/mesenteric node culture. - PCR for species/virulence genes (*inv*, *ypmA*, O-antigen gene clusters). - Imaging (CT/ultrasound): mesenteric lymphadenopathy, terminal ileitis — helps distinguish from surgical appendicitis.

Post-infectious / chronic disease — serology:

"Whereas early infections can be diagnosed by direct detection of bacteria, chronic infections can only be identified by serological tests." — Wielkoszynski 2018, [PMID: 30238343](#)

Species-discriminating serology uses recombinant antigens (F007):

"discrimination between the two types of infection is based on two recombinant bacterial proteins, MyfA and PsaA (specific for Y. enterocolitica and Y. pseudotuberculosis, respectively)" — [PMID: 30238343](#)

O-genotyping by multiplex PCR of O-antigen gene clusters replaces classical serotyping and distinguishes *Y. pseudotuberculosis* from *Y. pestis* (Bogdanovich 2003, [PMID: 14605146](#)).

Differential diagnosis: acute appendicitis, *Y. enterocolitica* infection, Crohn's terminal ileitis, Kawasaki disease (notably — several case reports link *Y. pseudotuberculosis* to KD-like presentations, e.g., [PMID: 9202805](#)), streptococcal scarlet fever, and other causes of reactive arthritis/erythema nodosum.

Omics diagnostics: not routine; PCR-based genotyping is the molecular standard.

11. Outcome / Prognosis

- **Overall prognosis: excellent.** Most infections are self-limiting with full recovery.
 - **Mortality:** low in immunocompetent hosts; significant risk of fatal **septicemia** in iron-overloaded, DFO-treated, splenectomized, or immunosuppressed patients (**F010**).
 - **Morbidity:** driven by post-infectious **reactive arthritis** (~12% of yersiniosis, weeks–months of disability) and **erythema nodosum** (~3%) (**F006**); rare myositis/myocarditis.
 - **Recovery:** high; antibiotics accelerate resolution of systemic disease.
 - **Prognostic factors:** host iron status, immune competence, and infecting-strain genotype (YPMa/pVM82 → severe FESLF). High **anti-YPM titers** correlate with systemic rather than purely GI disease (a prognostic serologic marker) (**F002**).
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12. Treatment

Pharmacotherapy. The organism is broadly antibiotic-susceptible (**F005**):

"None of the 58 Y. pseudotuberculosis isolates was resistant to any tested antimicrobial." — Kim 2017, [PMID: 28222842](#)

Drug class	Examples	Suggested NCIT	Notes
Fluoroquinolones	ciprofloxacin	NCIT:C540	First-line for systemic disease
3rd-gen cephalosporins	ceftriaxone, cefotaxime	NCIT:C1737	Used in systemic/severe cases (incl. KD-associated case)
Aminoglycosides	gentamicin, enrofloxacin (vet)	NCIT:C61796	Effective; note T3SS-associated tolerance in vitro
Tetracyclines	doxycycline, oxytetracycline	NCIT:C692	Effective; oxytetracycline used in livestock outbreaks
TMP-SMX	trimethoprim-sulfamethoxazole	NCIT:C287	Alternative (note sulfonamide resistance in some animal strains)

- **Mild self-limiting gastroenteritis** typically needs only supportive care (rehydration).
- **Systemic/FESLF/septicemia** and high-risk hosts warrant antibiotics.
- **Caveat — antibiotic tolerance:** T3SS-induced growth arrest can reduce susceptibility to gentamicin and doxycycline in subpopulations within host tissues ([PMID: 40623067](#); [PMID: 32753491](#)).

Supportive care: hydration; NSAIDs for reactive arthritis/erythema nodosum. **Surgical:** occasionally unnecessary appendectomy is performed before diagnosis is clarified. **No gene, cell, or RNA-based therapies** are relevant.

13. Prevention

No licensed human vaccine exists. Prevention is centered on interrupting foodborne transmission (**F012**):

- **Primary prevention (food/water hygiene):** wash raw produce; treat/boil water; safe meat handling; recognize that **refrigeration does not stop this psychrotroph**, so cold storage is not protective.
- **Secondary prevention:** early culture-based diagnosis to avoid unnecessary appendectomy and to guide antibiotics in high-risk hosts.

- **Tertiary prevention:** prompt antibiotics and iron-management vigilance in β -thalassemia/hemochromatosis/DFO patients to prevent septicemia (**F010**).
- **Public health / veterinary:** reservoir control in livestock and food-processing hygiene; molecular surveillance links animal/food strains to human cases (**F008**, [PMID: 19835774](#)).

14. Other Species / Natural Disease

Zoonotic pathogen with broad reservoirs (**F008**):

"Yersinia pseudotuberculosis and Yersinia enterocolitica are ubiquitous pathogens with wildlife and domestic animal reservoirs." — Walker 2018, [PMID: 30360909](#)

- **Natural disease in animals:** systemic *Y. pseudotuberculosis* O:1 causing multi-organ abscessation and osteomyelitis in a ring-tailed lemur (*Lemur catta*); outbreaks in zoological/research primate colonies (enteritis, mesenteric lymphadenitis, organ abscessation) (Walker 2018, [PMID: 30360909](#)).
- **Livestock:** serotype III causes "winter scours" diarrhea outbreaks in weaned Merino sheep, seasonally in winter (Stanger 2018, [PMID: 29691860](#)).
- **Wildlife surveillance:** beech marten and Alpine ibex identified as reservoirs, carrying mobilizable virulence plasmids (Carella 2022, [PMID: 35892105](#)).
- **Species affected (NCBI Taxonomy):** pigs (txid9823), sheep (txid9940), primates including *Lemur catta* (txid9447), deer, rodents, birds, humans (txid9606).

Comparative / evolutionary biology (F005**):** *Y. pseudotuberculosis* is the recent enteric ancestor of *Y. pestis*:

"Conventional microbiology, bacterial population genetics, and genome sequence data, all suggest that Y pestis is a recently evolved clone of the enteric pathogen Yersinia pseudotuberculosis." — Prentice & Rahalison 2007, [PMID: 17416264](#)

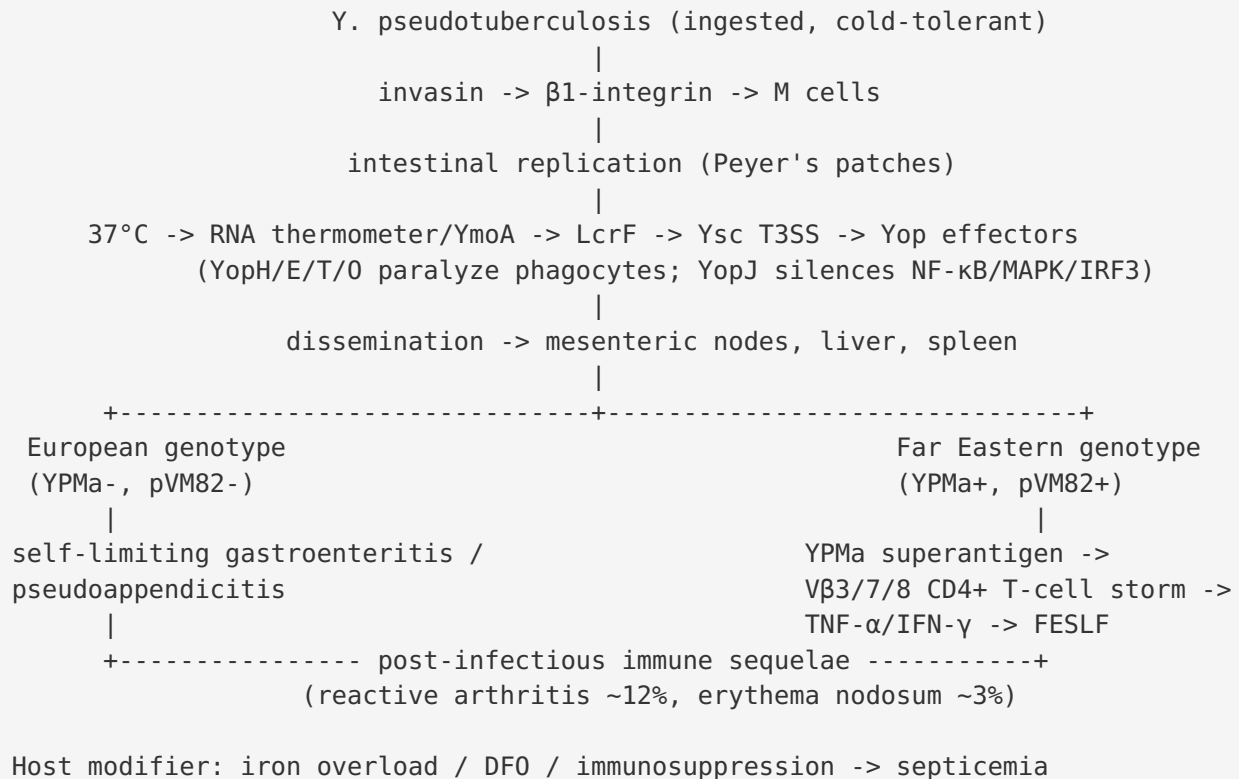
Zoonotic transmission: fecal–oral via contaminated food/water; strain identity between human, pig, sheep, and food isolates confirms cross-species/food-chain transmission (**F008**).

15. Model Organisms

- **Mouse (oral infection model)** — the principal model. Recapitulates M-cell/Peyer's-patch entry, intestinal replication, and hepatosplenic dissemination; used to prove the temperature/T3SS and superantigen mechanisms.
 - Böhme 2012 ([PMID: 22359501](#)): oral mouse infection demonstrated that a locked thermometer variant loses dissemination and lethality.
 - Barnes 2006 ([PMID: 16754724](#)): oral mouse model showed intestinal replication precedes dissemination.
 - Miyoshi-Akiyama 1997 ([PMID: 9159409](#)): murine YPM model established MHC-II-dependent V β 7/V β 8 expansion and TNF- α /IFN- γ -mediated lethal shock.
 - ***Galleria mellonella*** (invertebrate infection model) — used for virulence/regulation studies (e.g., Fis regulator, [PMID: 41880349](#)).
 - **Cell-culture / in vitro** — epithelial cell invasion assays (invasin- β 1-integrin), phagocyte intoxication assays (YopE/YopB-D), RNAi screens for host T3SS cofactors (e.g., CCR5, [PMID: 25691588](#)).
 - **Phenotype recapitulation:** the mouse oral model reproduces the enteric-to-systemic axis very well; it does **not** fully reproduce human FESLF because human-specific superantigen (YPMa) effects and the pVM82-linked epidemic phenotype are strain- and host-context dependent.
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Mechanistic Model / Interpretation

The disease is best understood as a **two-stage, genotype-branched infection**:



Upstream vs downstream: temperature sensing → LcrF → T3SS is the upstream master switch; Yop-mediated immune paralysis and (in FESLF) superantigen T-cell activation are downstream effectors; post-infectious arthritis/erythema nodosum are terminal immune sequelae. The pathogen genotype (plasmid/superantigen content) is the single strongest determinant of whether the outcome is mild or severe.

Evidence Base

PMID	Paper (abbreviated)	Supports	Contribution
26819960	Far East scarlet-like fever review (Amphlett 2016)	F001	Two clinical faces; YPMa in FESLF
9109426	Clinical role of superantigen (Abe 1997)	F002	Anti-YPM correlates with systemic disease
9159409	Murine YPM toxicity (Miyoshi-Akiyama 1997)	F002	V β /CD4, TNF- α /IFN- γ shock mechanism
22359501	RNA thermometer/LcrF (Böhme 2012)	F003	Temperature-gated virulence; in vivo proof
39159284	RNase reprogramming (Meyer 2024)	F003	T3SS-driven transcriptomic shift
12755365	Invasin/ β 1-integrin (Grassl 2003)	F004	M-cell entry mechanism
16754724	Intestinal dissemination (Barnes 2006)	F004	Replication precedes dissemination
17416264	Plague (Prentice 2007)	F005	Ancestor of <i>Y. pestis</i>
14605146	O-genotyping PCR (Bogdanovich 2003)	F005	≥ 21 serotypes; O:1b origin of <i>Y. pestis</i>
28222842	AMR in primates (Kim 2017)	F005	Broad antibiotic susceptibility
33321204	Pan-genomics/drug mining (Basharat 2021)	F006	Phenotype spectrum enumeration
23701958	<i>Yersinia sequeleae</i> (Rosner 2013)	F006	ReA ~12%, EN ~3% frequencies
30238343	Diagnostic ELISA (Wielkoszynski 2018)	F007	Culture vs serology; PsaA marker
30360909	Lemur osteomyelitis (Walker 2018)	F008	Zoonotic reservoir; animal disease
19835774	Nigeria human/animal strains (Okwori 2009)	F008	Food/animal-to-human strain identity
29691860	Winter scours in sheep (Stanger 2018)	F008	Livestock natural disease
17608743	YopJ/TRAF (Sweet 2007)	F009	NF- κ B/MAPK/IRF3 inhibition

PMID	Paper (abbreviated)	Supports	Contribution
11132234	Infections in E- β thalassemia (Wanachiwanawin 2000)	F010	Iron/DFO risk factor
30695393	Plasmid-associated virulence (Somova 2016)	F011	pVM82 + YPMa $\rightarrow$ FESLF
32498317	FESLF as special manifestation (Somova 2020)	F011	Epidemiologic geographic pattern
20088683	Cold adaptation (Palonen 2010)	F012	Psychrotrophic foodborne transmission

Limitations and Knowledge Gaps

- 1. Quantitative human epidemiology is sparse.** Precise incidence/prevalence (per 100,000) for *Y. pseudotuberculosis* specifically are not well established; many sequelae frequencies (reactive arthritis ~12%, erythema nodosum ~3%) are **extrapolated from *Y. enterocolitica*** population studies rather than measured for *Y. pseudotuberculosis* directly.
- 2. FESLF molecular detail is understudied.** The pVM82 plasmid's gene content and the precise mechanistic link between pVM82 and the FESLF phenotype remain incompletely defined; YPMa's role is strongly implicated but the causal chain to renal dysfunction is partly inferred.
- 3. No human host-genetic risk data.** Because this is infectious, no GWAS/germline susceptibility data exist; iron availability is the dominant, well-supported host modifier.
- 4. Kawasaki-disease association is unresolved.** Several case reports link *Y. pseudotuberculosis* to KD-like illness, but causation vs. coincidence/molecular mimicry is not established.
- 5. Antibiotic tolerance in tissue** (T3SS-induced growth arrest reducing aminoglycoside/tetracycline killing) is demonstrated in mouse models but its clinical treatment-failure significance in humans is unquantified.
- 6. No original data analysis** was performed in this investigation; findings are literature-derived syntheses, so confidence rests on the quality of the cited primary studies.

Proposed Follow-up Experiments / Actions

1. **Quantify *Y. pseudotuberculosis*-specific sequelae** in a prospective human cohort (rather than extrapolating from *Y. enterocolitica*), stratified by infecting serotype and YPMa/pVM82 status.
2. **Complete pVM82 functional genomics:** sequence and mutate pVM82 loci to define which genes are necessary/sufficient for FESLF-associated renal and vascular pathology.
3. **Structure-guided anti-superantigen therapeutics:** build on the YPM point-mutant work ([PMID: 10087177](#)) to develop TCR-V β -blocking or attenuated-toxoid immunotherapeutics for severe FESLF.
4. **Clinical tolerance study:** test whether T3SS-induced antibiotic tolerance drives relapse in systemic human disease and whether combination regimens or T3SS inhibitors improve clearance.
5. **Iron-management guideline evaluation:** assess whether temporary DFO cessation and prompt empirical antibiotics reduce septicemia mortality in iron-overloaded patients presenting with fever.
6. **Integrated One-Health surveillance:** expand PFGE/WGS matching of food, livestock, wildlife, and human isolates to map transmission chains and identify high-risk food vehicles for targeted prevention.

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